

Harvard (HEEDB) EEG labels — how we read the fields we compare against

Before trusting any comparison, we had to be precise about what Harvard's label columns actually mean. This note is that: what the fields are, how each cell is coded, and — for the specific fields we compare our MedGemma labels against — the exact definition, how it maps to our field, and how common it is. It is verified from the data-descriptor paper's supplement (Sun et al., *Epilepsia* 2025, **Data S1**) and checked against the database itself.

The tables

Each Harvard hospital has one table, <COHORT>_EEG_with_reports, with a row per recording session and one column per finding (39 findings) alongside identifier/metadata columns. A report (DeidentifiedName(Reports)) spans several session rows, so we work at the report level (present if any of its rows carry the finding).

How a cell is coded — the part that's easy to get wrong

A finding cell is **not** a yes/no. It is a short **provenance string** saying *how* the finding was established, or it is empty. Harvard derived findings three ways (Data S1 §A; paper abstract):

- **annotation** — keyword / regular-expression matching over the recording's EEG annotations (Data S1 Table S5).
- **report** — a fine-tuned medical LLM reading the free-text report and answering a yes / no / not-mentioned question per finding. Naming: Data S1 §A.2 calls it *Medical LLaMA 3-8B* (the paper's own text is inconsistent on the size, also writing "Llama 3 7B"); the public tooling (process_reports_by_medicalllama.py and the repo README) calls it *Medical-LLaMA* and links it to the Hugging Face model ContactDoctor/Bio-Medical-Llama-3-8B — the three names are the same model. The production prompt template is given in Data S1: *"Based on the medical context of neurology and EEG analysis, does the following report indicate the presence of [finding] in the EEG? Please respond with yes, no, or not mentioned."* (The example script in the public repo uses a simpler seizure-specific "Answer Yes or No" phrasing — a demo, not the full 33-finding template.)
- **verified** — checked by a person (expert curation).

A cell holds whichever applied, combined: report, annotation, verified, report annotation, report verified, report annotation verified, annotation verified, or empty. So the column is a **merge of both extraction channels plus verification**, not a single channel. (An earlier version of this note called the column "the report channel" — that was wrong; the report channel is only the subset of cells whose tag contains report.)

Two consequences we rely on:

- **non-null = present; empty = not recorded.** There is no value meaning "absent", so empty bundles three different things — the report said absent, it wasn't mentioned, or nobody annotated it. Harvard can't separate them; we keep explicitly_absent vs not_mentioned on our side but collapse to present-vs-not to compare. Data S1 §A.4 is explicit about this: *"'not mentioned' and 'not available' are common, but they do not necessarily indicate a negative finding. Therefore ... we can only assess the agreement between the two when the finding is marked as 'yes'."*
- **the tag tells you the source**, which matters because we read only the report text. Comparing against report-tagged cells is the fair like-for-like; annotation-only cells may describe something never written in the narrative.

Exactly what we count as "present" in each comparison

A cell is a set of up to three tags — report, annotation, verified — or empty. Each of the three comparisons in our reports reads the cell differently:

cell value	meaning	“vs all” (κ_{all})	“vs report-text” (κ_{report})	in verified set (recall)
(empty)	not recorded (absent / not-mentioned / not-annotated)	not present	not present	no
report	LLM found it in the report text	present	present	no
annotation	regex found it in the recording annotations	present	not present	no
verified (alone)	expert-flagged, no channel tag (rare)	present	not present	yes
report annotation	both channels	present	present	no
report verified	in the text + expert-verified	present	present	yes
annotation verified	in annotations + expert-verified	present	not present	yes
report annotation verified	both channels + expert-verified	present	present	yes

So: **“vs all”** counts any non-empty cell as present; **“vs report-text”** counts a cell as present only if its tag contains report (the fair like-for-like, since we read only the text); the **verified set** is every cell whose tag contains verified (recall = how many of those we also called present). Our own side is always collapsed to present vs not-present (explicitly_absent and not_mentioned both count as not-present). A report spans several session rows, so it is present-in-text if any of its rows carries a report tag.

The present-set at each level, written out. Against Harvard, “present” is exactly:

- **vs all** — report, annotation, verified, report annotation, report verified, annotation verified, report annotation verified (every non-empty cell); only empty is not-present.
- **vs report-text** — the cells whose tag contains report: report, report annotation, report verified, report annotation verified. Here annotation-only, verified-only and annotation verified count as not-present, alongside empty.
- **verified set** — the cells whose tag contains verified: verified, report verified, annotation verified, report annotation verified. Positives only (there is no “verified absent”), so we report recall on this set, not κ .

What verified actually confirmed. It is a separate, stronger step than the report itself. Every report label already comes from a clinical report a clinician wrote and signed (Data S1 §A.4: “reviewed and confirmed by at least two clinicians, including a senior clinician”). The verified tag sits on top of that — Harvard’s *multi-expert* labels, which Data S1 §A.4 calls “event-level ground truth”, the strongest of the three tiers. An expert confirmed that the specific finding is genuinely present in that recording, independent of what the report LLM or the annotation regex said. It is positives-only (no expert “absent”) and per-finding, not per-report. It can even stand alone — e.g. MGH foc slowing carries 1,262 cells tagged verified with no report or annotation: present by expert review although neither automated channel flagged it.

We confirmed this on the data rather than assuming it:

check	result
report-tagged cells whose report text contains the finding word	100%
annotation-only cells whose report text contains it	72% (often not in the narrative)

check	result
foc slowing and gen slowing verified in the same MGH report	49 — so verification is per-field, not per-report

Cross-check against the public GitHub label tables

The same label tables are published in `bdsp-core/Harvard-EEG-Database-Tools` (MGH/BWH/BCH/BIDMC_EEG_with). We downloaded them and compared them column by column against the `HarvardEEG.db` we query: the row counts and the full value vocabulary match **exactly**, cell for cell. For MGH bs, both sides give 75,994 empty / 10,930 report / 2,162 annotation / 1,199 report annotation / 201 report annotation verified / 159 report verified / 50 annotation verified / 21 verified. So the database we compare against is the same public source, and the eight values above are the complete cell vocabulary — there is no ninth code we might have missed.

The fields we actually compare against

These are the Harvard columns we line our labels up with, with report-level present rate per hospital and their Table S5 annotation keyword. The present rate here is over **all** of that hospital's reports; the comparison reports instead report it over the reports we and Harvard both labeled (the intersection), so the same figure comes out a point or two lower there — e.g. bs is 12% of all MGH reports but 11% of the intersection:

Harvard column	our field	present — MGH / BWH / BCH	Table S5 keyword	notes
abnormal	abnormality	66% / 67% / 58%	abnormal	clean 1:1; 100% report-derived (LLM only, no annotation/verified cells)
foc slowing	focal_slowing	14% / 24% / 7%	foc* slow*, r* slow*, l* slow*	just the word “slowing” with laterality — narrower than our old focal <i>non-epileptiform</i> bucket
gen slowing	generalized_slowing	77% / 72% / 27%	gen* slow*	just “generalized slowing” keyword also matches plain “suppression” , so the annotation channel is broad — a big reason Harvard's bs is marked 3-6× more than our text-only reading (at MGH/BWH; at BCH their bs is mostly text-derived)
bs	burst_suppression	12% / 15% / 9%	burst, suppression, supp, bsr	

Two of these were also touched by earlier work: abnormal is the one clean field in the abnormality comparison, and foc/gen slowing were the approximate targets in the older focal/generalized *non-*

epileptiform mapping (our non-epi bucket lumps slowing together with attenuation, asymmetry, disorganization and excessive beta, so it is broader than Harvard's slowing-only column — their own Table S5 keyword confirms the mismatch, not just our analysis/mapping_check.py).

Triphasic waves, discontinuous background and suppressed background have **no** Harvard column at all (nearest to suppressed is low voltage, a different thing), so those stay labels-only.

All 39 finding fields

Meaning is the standard-EEG reading of the column name; the keyword column is Harvard's actual Table S5 regex for the **annotation** channel (the report channel used the LLM question above, not these keywords), shown because it is their published definition and reveals how narrow some fields are.

#	Field	What it means (standard EEG terminology)	Annotation keyword (Data S1 Table S5)
1	normal	EEG reported normal / within normal limits	normal
2	abnormal	EEG reported abnormal (any abnormal finding)	abnormal
3	awake	Awake state captured	eo, awake
4	n1	Sleep stage N1	drowsiness, drowsy, n1
5	n2	Sleep stage N2	spindle, spindles, ss, kss, n2, ii
6	bets	Benign epileptiform transients of sleep (small sharp spikes)	bets, sss
7	wicket	Wicket spikes (benign variant)	wicket[s]
8	spindles	Sleep spindles	spindle, spindles, ss, kss
9	vertex wave	Vertex sharp transients	v, vw, vert, vertw, vertwave, vertex, vertexwave
10	posts	Positive occipital sharp transients of sleep	posts
11	breach	Breach rhythm (activity over a skull defect)	bre[ea]ch
12	spikes	Epileptiform spikes / sharp waves	spike[s], discharge[s], sharp
13	seizure	Electrographic / clinical seizure	sz, seizure[s]
14	lpd	Lateralized periodic discharges	lp[e]d[s], pled
15	gpd	Generalized periodic discharges	gpd[s]
16	lrda	Lateralized rhythmic delta activity	lrda, tirda
17	grda	Generalized rhythmic delta activity	grda, [of]irda
18	bs	Burst-suppression	burst, suppression, supp, bsr
19	foc slowing	Focal slowing	foc* slow*, r* slow*, l* slow*, l* temporal slow*
20	gen slowing	Generalized / diffuse slowing	gen* slow*

#	Field	What it means (standard EEG terminology)	Annotation keyword (Data S1 Table S5)
21	pdr	Posterior dominant rhythm (normal background)	pdr
22	dravet	Dravet syndrome pattern	<i>(not in Data S1 — undocumented)</i>
23	eses	Electrical status epilepticus in sleep	eses
24	fold	Unclear — appears in Data S1 figures but has no keyword or definition	<i>(none)</i>
25	jae	Juvenile absence epilepsy	jea
26	jeavons	Jeavons syndrome	<i>(not in Data S1 — undocumented)</i>
27	jme	Juvenile myoclonic epilepsy	hme
28	k_complexes	K-complexes	k-complexes, k
29	bects	Benign epilepsy with centrotemporal spikes	bects
30	bipd	Bilateral independent periodic discharges	bipd
31	cjd	Creutzfeldt-Jakob periodic pattern	cjd
32	diffuse Beta	Diffuse / excessive beta	diffuse beta
33	low voltage	Low-voltage background	low voltage
34	ppr	Photoparoxysmal response	ppr
35	status	Status epilepticus	<i>(listed as a finding, no keyword given)</i>
36	sunflower	Sunflower syndrome	<i>(not in Data S1 — undocumented)</i>
37	uninterpretable	Recording uninterpretable	<i>(in figures, no keyword)</i>
38	wham	Unclear — not standard EEG terminology, absent from Data S1	<i>(none)</i>
39	angelman	Angelman syndrome pattern	angelman

The keyword lists carry small typos (seizure serizure, JAE jea, JME hme) — those affect only the annotation channel, not the report labels. Six columns (dravet, fold, jeavons, sunflower, uninterpretable, wham) fall outside the paper’s 33 documented findings, so they are effectively undocumented.

The identifier / metadata columns: SiteID, BDSPPatientID, SessionID, AgeAtVisit, SexDSC, Service-Name(EEG), CreationTime(EEG), StartTime(EEG), EndTime(EEG), ProcedureDSC(Reports), EncounterDTS(Reports), BeginDTS(Reports), ExamEndDTS(Reports), DeidentifiedName(Reports).

Reliability (Data S1 §A.4)

There is no gold standard, so Harvard reports an “ambiguity” rate — the share of true-yes cases the report LLM missed (called no): **2.25%** on average against annotations, and **0%** against multi-expert labels (the LLM never said no where experts said yes). So the report channel is high-recall and rarely

denies a finding that is present. The report labels were also “reviewed and confirmed by at least two clinicians, including a senior clinician.”

Who read each report

There is no author column. The reading physician appears only inside the report text, as a signature — e.g. “*I (... , MD) have reviewed the study and edited this report.*” Reports are de-identified with **PHILter** (rule-based + statistical PHI removal, HIPAA Safe Harbor, with date shifting), which is why most signatures collapse to ***** while a few readers’ initials slip through:

Hospital	Reports	Has “I (...)” signature	Any attestation phrase	Distinguishable reader
MGH	55,557	35,652 (64%)	54,233 (98%)	8,182
BWH	21,035	2 (0%)	11,619 (55%)	2
BCH	25,022	0 (0%)	7,297 (29%)	0
Total	101,614	35,654 (35%)	73,149 (72%)	8,184 (8%)

The distinguishable readers (≈ 14 , almost all MGH: ADL, ESR, LM, DBH, JS, CMM, LTG, KCS, ...) are in `results/harvard/report_authors.csv`.

Sources

- Sun et al., **Harvard Electroencephalography Database: A comprehensive clinical electroencephalographic resource from four Boston hospitals**, *Epilepsia* 2025 — doi:10.1111/epi.18487 — <https://onlinelibrary.wiley.com/doi/10.1111/epi.18487> (§A extraction method + LLM prompt, Table S5 keywords, Table S2 finding counts, §A.4 reliability, §D de-identification — all in the **Data S1** supplement).
- Dataset (credentialed), Harvard EEG Database v4.1 — <https://bdsp.io/content/harvard-eeg-db/4.1/>
- Extraction tooling and shared files — `bdsp-core/Harvard-EEG-Database-Tools` — <https://github.com/bdsp-core/Harvard-EEG-Database-Tools>. The public repo holds the label tables themselves (MGH/BWH/BCH/BIDMC_EEG_with_reports.csv.zip), the LLM extraction example (`process_reports_by_medicalllama.py`, `format_reports.py` — seizure only), the ICD-10 and ATC medication dictionaries, `HEEDB_patients.csv`, and the statistics notebooks (`read_data.ipynb`, `HEEDB_statistics.ipynb`). It does **not** publish the full per-finding prompts or the keyword definitions beyond the seizure example — those are only in Data S1.
- Our own checks: `analysis/slowing_vs_harvard.py` and `analysis/background_vs_harvard.py` (provenance handling), `analysis/mapping_check.py` (the slowing-vs-non-epi mismatch); label schema in `HarvardEEG.db`; reader signatures in `results/harvard/report_authors.csv`.